Stat 654 – Homework 5

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Problem 1 (Gaussian couplings)**.** *Recall that a coupling of two random variables with given marginals means a joint construction of them on the same probability space. Let $X, Y \sim N(0,1)$ be two independent $N(0,1)$ random variables.*

1. *Let $Z := aX + bY$. You may use without proof that Z is also Gaussian. Compute the mean and variance of Z. For what condition on a and b does Z have distribution $N(0,1)$?*

   Solution. *Let $Z := aX + bY$ with $X, Y \sim N(0,1)$ independent. Then*

   $$\mathbb{E}[Z] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] = 0.$$

   Also,

   $$\mathrm{Var}(Z) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y).$$

   Since X and Y are independent, $\mathrm{Cov}(X,Y) = 0$, *and* $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$, *we get*

   $$\mathrm{Var}(Z) = a^2 + b^2.$$

   Therefore $Z \sim N(0,1)$ if and only if $a^2 + b^2 = 1$.

2. *Show that $(X, \rho X + \sqrt{1-\rho^2}Y)$ is a coupling of two standard Gaussian random variables for any $\rho \in [-1,1]$. Draw a scatterplot of the pair and plot histograms of both coordinates to numerically verify that each marginal distribution is standard Gaussian. This coupling is called the identical coupling when $\rho = 1$, the independent coupling when $\rho = 0$, and the antithetic coupling when $\rho = -1$.*

   Solution. *Define*

   $$Z_\rho := \rho X + \sqrt{1-\rho^2}\,Y, \qquad \rho \in [-1,1].$$

   Then

   $$\mathbb{E}[Z_\rho] = \rho\,\mathbb{E}[X] + \sqrt{1-\rho^2}\,\mathbb{E}[Y] = 0,$$

   and

   $$\mathrm{Var}(Z_\rho) = \rho^2\,\mathrm{Var}(X) + (1-\rho^2)\,\mathrm{Var}(Y) + 2\rho\sqrt{1-\rho^2}\,\mathrm{Cov}(X,Y) = \rho^2 + (1-\rho^2) = 1,$$

   since X and Y are independent with $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$ *and* $\mathrm{Cov}(X,Y) = 0$. *Hence $Z_\rho \sim N(0,1)$, so (X, Z_ρ) is a coupling of two standard Gaussian random variables for every $\rho \in [-1,1]$.*

   For a numerical check, I simulated $n = 10^5$ i.i.d. pairs (X_i, Y_i) from $N(0,1)$ and formed

   $$Z_{\rho,i} = \rho X_i + \sqrt{1-\rho^2}\,Y_i, \qquad i = 1,\ldots,n,$$

   for $\rho \in \{1, 0, -1\}$. A short Python script produced scatterplots of $(X_i, Z_{\rho,i})$ and overlaid histograms of the marginals X and Z_ρ; the resulting figures are included below.

   In all three cases, the histograms of X and Z_ρ look almost identical and match the bell shape of the standard normal density, so both marginals are close to $N(0,1)$. The scatterplots have sample correlations close to ρ, so we see the coupling move from identical ($\rho = 1$) to independent ($\rho = 0$) to antithetic ($\rho = -1$).

3. *Consider $(X_1, X_2) = (aX + bY, cX + dY)$. Find values of (a, b, c, d) such that (a) (X_1, X_2) is a coupling of two standard Gaussian random variables, and (b) $X_1 + X_2 = X$.*

   Solution. *Let*
$$X_1 = aX + bY, \qquad X_2 = cX + dY.$$
   Since X and Y are independent standard normals, requiring X_1 and X_2 to be standard Gaussian gives
$$a^2 + b^2 = 1, \qquad c^2 + d^2 = 1.$$
   The condition $X_1 + X_2 = X$ implies
$$(a + c)X + (b + d)Y = X,$$
   so
$$a + c = 1, \qquad b + d = 0.$$
   One convenient family of solutions is obtained as follows: choose any $b \in [-1, 1]$ and set
$$a = \sqrt{1 - b^2}, \qquad c = 1 - a, \qquad d = -b.$$
   Then
$$a^2 + b^2 = 1,$$
   and also
$$a + c = a + (1 - a) = 1, \qquad b + d = b - b = 0.$$
   Finally,
$$c^2 + d^2 = (1 - a)^2 + b^2 = 1 - 2a + a^2 + b^2 = 1,$$
   because $a^2 + b^2 = 1$. Hence this choice gives a valid coupling of two standard Gaussian random variables satisfying $X_1 + X_2 = X$.

Problem 2 (Random walk on the complete graph)**.** *Consider the Markov chain on $\Omega = \{1, 2, \ldots, n\}$ defined as follows: if the chain is currently at state x, then it moves uniformly to one of the other $n - 1$ states. In other words,*
$$P(x, z) = \frac{1}{n-1}\,\mathbf{1}\{z \neq x\}.$$

1. *Show that the uniform distribution on Ω is stationary for this Markov chain.*

   Solution. *Let π be the uniform distribution on Ω, so $\pi(x) = 1/n$ for each x. For any $z \in \Omega$,*
$$(\pi P)(z) = \sum_{x\in\Omega} \pi(x)P(x, z) = \sum_{x\neq z} \frac{1}{n} \cdot \frac{1}{n-1} = \frac{n-1}{n} \cdot \frac{1}{n-1} = \frac{1}{n} = \pi(z).$$
   Therefore the uniform distribution is stationary.

2. *Let $x \neq y$. Compute $\|P(x,\cdot) - P(y,\cdot)\|_{\mathrm{TV}}$.*

   Solution. *By definition,*

$$\|P(x,\cdot) - P(y,\cdot)\|_{\mathrm{TV}} = \frac{1}{2}\sum_{z\in\Omega}\big|P(x,z) - P(y,z)\big|.$$

   The two distributions differ only at $z = x$ and $z = y$. Indeed,

$$|P(x,x) - P(y,x)| = \left|0 - \frac{1}{n-1}\right| = \frac{1}{n-1},$$

   and

$$|P(x,y) - P(y,y)| = \left|\frac{1}{n-1} - 0\right| = \frac{1}{n-1}.$$

   For all other z we have $P(x,z) = P(y,z) = 1/(n-1)$, so those terms contribute zero. Thus

$$\|P(x,\cdot) - P(y,\cdot)\|_{\mathrm{TV}} = \frac{1}{2}\left(\frac{1}{n-1} + \frac{1}{n-1}\right) = \frac{1}{n-1}.$$

3. *Construct a coupling (X_1, Y_1) of $P(x,\cdot)$ and $P(y,\cdot)$ such that*

$$\mathbb{P}(X_1 \neq Y_1) = \|P(x,\cdot) - P(y,\cdot)\|_{\mathrm{TV}}.$$

   Solution. *Define a coupling of $P(x,\cdot)$ and $P(y,\cdot)$ as follows.*

   - *With probability $\frac{1}{n-1}$, set $(X_1, Y_1) = (y, x)$.*
   - *With the remaining probability $1 - \frac{1}{n-1}$, choose Z uniformly from $\Omega \setminus \{x, y\}$ and set $(X_1, Y_1) = (Z, Z)$.*

   We first check the marginals. For the X–coordinate,

$$\mathbb{P}(X_1 = y) = \frac{1}{n-1}, \qquad \mathbb{P}(X_1 = x) = 0,$$

   and for any $z \in \Omega \setminus \{x, y\}$,

$$\mathbb{P}(X_1 = z) = \left(1 - \frac{1}{n-1}\right)\mathbb{P}(Z = z) = \frac{n-2}{n-1}\cdot\frac{1}{n-2} = \frac{1}{n-1}.$$

   So X_1 is uniform on $\Omega \setminus \{x\}$, i.e. $X_1 \sim P(x,\cdot)$. The same calculation with x and y swapped shows $Y_1 \sim P(y,\cdot)$.

   Under this coupling the chains disagree only in the first case, when $(X_1, Y_1) = (y, x)$. Hence

$$\mathbb{P}(X_1 \neq Y_1) = \frac{1}{n-1}.$$

   From part (2), $\|P(x,\cdot) - P(y,\cdot)\|_{\mathrm{TV}} = 1/(n-1)$, so this coupling achieves the total variation distance.

4. *Now run this coupling at every step. Let*

$$T := \inf\{t \geq 0 : X_t = Y_t\}.$$

Show that if $X_0 \neq Y_0$, then

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

Solution. *We run at every step the coupling from part (3), and define*

$$T := \inf\{t \geq 0 : X_t = Y_t\}.$$

I claim that for $x \neq y$,

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

Whenever $X_t \neq Y_t$, the one–step coupling satisfies

$$\mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t) = \frac{1}{n-1}.$$

Moreover, once the chains meet, the coupling keeps them together forever: if $X_t = Y_t$ then $X_{t'} = Y_{t'}$ for all $t' \geq t$.

For $t = 0$ and $x \neq y$ we have $T > 0$ almost surely, so

$$\mathbb{P}(T > 0) = 1 = (1/(n-1))^0.$$

Now suppose the formula holds for some $t \geq 0$. Using the Markov property of the coupled chain and the fact that $T > t$ is equivalent to $X_t \neq Y_t$,

$$\mathbb{P}(T > t+1) = \mathbb{P}(X_{t+1} \neq Y_{t+1} \mid X_t \neq Y_t)\,\mathbb{P}(T > t) = \frac{1}{n-1}\left(\frac{1}{n-1}\right)^t = \left(\frac{1}{n-1}\right)^{t+1}.$$

This proves the claimed formula.

5. *Use the coupling inequality to conclude that*

$$\max_{x,y} \|P^t(x,\cdot) - P^t(y,\cdot)\|_{\mathrm{TV}} \leq \left(\frac{1}{n-1}\right)^t.$$

Solution. *By the coupling inequality,*

$$\|P^t(x,\cdot) - P^t(y,\cdot)\|_{\mathrm{TV}} \leq \mathbb{P}(T > t).$$

Using the coupling from the previous parts,

$$\mathbb{P}(T > t) = \left(\frac{1}{n-1}\right)^t.$$

Hence

$$\max_{x,y} \|P^t(x,\cdot) - P^t(y,\cdot)\|_{\mathrm{TV}} \leq \left(\frac{1}{n-1}\right)^t.$$

Problem 3 (Random-to-top shuffle). *Consider a deck of n cards labeled $1, \dots, n$. A shuffling is a permutation of $\{1, 2, \dots, n\}$. The random-to-top shuffle is the Markov chain on permutations where at each step we choose a card label uniformly from $\{1, \dots, n\}$, remove that card from its current position, and move it to the top of the deck.*

1. *Show that the uniform distribution on all $n!$ permutations is stationary for this Markov chain.*

   Solution. *Let π be the uniform distribution on S_n, so $\pi(\sigma) = 1/n!$ for each $\sigma \in S_n$. Fix $\tau \in S_n$. A transition $\sigma \to \tau$ is possible exactly when σ is obtained by taking τ and inserting its top card back into one of the n positions in the deck. There are n such permutations $\sigma_1, \dots, \sigma_n$, and for each*
   $$P(\sigma_k, \tau) = \frac{1}{n},$$
   since we choose that card label with probability $1/n$. Therefore
   $$(\pi P)(\tau) = \sum_{k=1}^{n} \pi(\sigma_k) P(\sigma_k, \tau) = n \cdot \frac{1}{n!} \cdot \frac{1}{n} = \frac{1}{n!} = \pi(\tau).$$
   Thus π is stationary for the random-to-top shuffle.

2. *Let X_t and Y_t be two copies of the random-to-top shuffle, started from (possibly) different initial decks X_0 and Y_0. Couple them by choosing the same card label at every step and moving that label to the top in both decks.*

   Let A_t be the set of card labels that have been chosen at least once by time t. Show that the relative order of the cards in A_t is the same in X_t and Y_t.

   Solution. *We prove the claim by induction on t. For $t = 0$, the set A_0 is empty, so the statement is trivial.*

   Assume the claim holds at time t, and consider time $t+1$. Let i_{t+1} be the label chosen at step $t+1$.

   Case 1: $i_{t+1} \in A_t$. Then in both decks we move the same card i_{t+1} to the top. In each deck, the positions of all other labels in A_t do not change; only i_{t+1} moves above them. Since the relative order of the labels in A_t was the same in X_t and Y_t by the induction hypothesis, and we apply the same operation to i_{t+1} in both decks, the relative order of the labels in A_t remains the same in X_{t+1} and Y_{t+1}. In this case $A_{t+1} = A_t$.

   Case 2: $i_{t+1} \notin A_t$. Then $A_{t+1} = A_t \cup \{i_{t+1}\}$, and in both decks the new label i_{t+1} is moved to the top of the deck. In particular, i_{t+1} is placed above all labels in A_t in both X_{t+1} and Y_{t+1}, while the relative order of the labels already in A_t is unchanged. Thus the relative order of the labels in A_{t+1} is the same in X_{t+1} and Y_{t+1}.

   In either case, the relative order of the labels in A_{t+1} is the same in the two decks. By induction, the claim holds for all t.

3. *Show that if $A_t = \{1, \dots, n\}$, then $X_t = Y_t$. Let*
   $$T := \inf\{t \geq 0 : X_t = Y_t\}.$$
   Deduce that
   $$\mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}).$$

Solution. *If $A_t = \{1, \dots, n\}$, then every label has been selected at least once. By the previous part, the relative order of all labels is the same in X_t and Y_t, so $X_t = Y_t$.*

Therefore, if $T > t$, then necessarily $A_t \neq \{1, \dots, n\}$. Hence

$$\mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}).$$

4. *Use the union bound to prove that*

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq n\left(1 - \frac{1}{n}\right)^t \leq ne^{-t/n}.$$

Solution. *For each label i, let E_i be the event that label i has not been chosen by time t. Then*

$$\mathbb{P}(E_i) = \left(1 - \frac{1}{n}\right)^t.$$

If $A_t \neq \{1, \dots, n\}$, then at least one E_i occurs. By the union bound,

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq \sum_{i=1}^{n} \mathbb{P}(E_i) = n\left(1 - \frac{1}{n}\right)^t.$$

Using $1 - 1/n \leq e^{-1/n}$ gives

$$\mathbb{P}(A_t \neq \{1, \dots, n\}) \leq ne^{-t/n}.$$

5. *Using the coupling inequality, conclude that*

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\mathrm{TV}} \leq ne^{-t/n}.$$

In particular, show that the mixing time is at most

$$t_{\mathrm{mix}}(\varepsilon) \leq n \log n + n \log(1/\varepsilon).$$

Solution. *By the coupling inequality and the previous bounds,*

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\mathrm{TV}} \leq \mathbb{P}(T > t) \leq \mathbb{P}(A_t \neq \{1, \dots, n\}) \leq ne^{-t/n}.$$

To make this at most ε, it suffices that $ne^{-t/n} \leq \varepsilon$. Taking logarithms yields

$$t \geq n \log n + n \log(1/\varepsilon).$$

Therefore,

$$t_{\mathrm{mix}}(\varepsilon) \leq n \log n + n \log(1/\varepsilon).$$

Problem 4 (Conditional Bernoulli distribution)**.** *Fix integers* $0 < I < N$ *and parameters* $(p_1, \ldots, p_N) \in (0,1)^N$*. The conditional Bernoulli distribution is the distribution on* $\{0,1\}^N$ *defined by*

$$\mathbb{P}(X_1 = x_1, \ldots, X_N = x_N) \propto \prod_{k=1}^{N} p_k^{x_k}(1-p_k)^{1-x_k}\mathbf{1}\left(\sum_{i=1}^{N} x_i = I\right).$$

Equivalently, $(X_1, \ldots, X_N)$ *are independent* $\mathrm{Ber}(p_k)$ *variables conditioned on* $\sum_{i=1}^{N} X_i = I$*.*

We consider the following MCMC algorithm. Given a current state $x^{(t)} \in \{0,1\}^N$ *with* $\sum_i x_i^{(t)} = I$*, define*

$$S_0^{(t)} := \{i : x_i^{(t)} = 0\}, \qquad S_1^{(t)} := \{j : x_j^{(t)} = 1\}.$$

At each iteration, choose $i_t \in S_0^{(t)}$ *and* $j_t \in S_1^{(t)}$ *independently and uniformly at random. The proposal flips* $x_{i_t}^{(t)}$ *and* $x_{j_t}^{(t)}$ *and is then accepted or rejected according to the Metropolis– Hastings rule.*

1. *Calculate the acceptance probability for a proposed flip of coordinates* i_t *and* j_t*.*

   Solution. *Let* $x \in \mathcal{X}$ *be the current state and let* y *be obtained by changing* $x_i = 0$ *to* $y_i = 1$ *and* $x_j = 1$ *to* $y_j = 0$*, for some* $i \in S_0(x)$ *and* $j \in S_1(x)$*. Since* $|S_0(x)| = N - I$ *and* $|S_1(x)| = I$*, the proposal probability is*

   $$q(x,y) = \frac{1}{|S_0(x)|}\,\frac{1}{|S_1(x)|} = \frac{1}{(N-I)I}.$$

   The reverse move $y \to x$ *has the same proposal probability, so* $q(y,x) = q(x,y)$*. The proposal is therefore symmetric, and the MH acceptance probability is*

   $$\alpha(x,y) = \min\Big\{1,\ \frac{\pi(y)}{\pi(x)}\Big\}.$$

   Since π *is only defined up to a normalizing constant, we can write*

   $$\pi(x) \propto \prod_{k=1}^{N} p_k^{x_k}(1-p_k)^{1-x_k}$$

   on $\mathcal{X}$*, and the constant drops out in the ratio. Only coordinates* i *and* j *change, so*

   $$\frac{\pi(y)}{\pi(x)} = \frac{p_i^{y_i}(1-p_i)^{1-y_i}p_j^{y_j}(1-p_j)^{1-y_j}}{p_i^{x_i}(1-p_i)^{1-x_i}p_j^{x_j}(1-p_j)^{1-x_j}} = \frac{p_i}{1-p_i}\cdot\frac{1-p_j}{p_j}.$$

   Thus

   $$\alpha(x,y) = \min\left\{1,\ \frac{p_i}{1-p_i}\cdot\frac{1-p_j}{p_j}\right\}.$$

2. *Prove that the Markov chain is irreducible on*

   $$\mathcal{X} := \Big\{x \in \{0,1\}^N : \sum_{i=1}^{N} x_i = I\Big\}.$$

Solution. *Take any $x, y \in \mathcal{X}$. Let*

$$A = \{k : x_k = 1, y_k = 0\}, \qquad B = \{k : x_k = 0, y_k = 1\}.$$

Since both x and y have exactly I ones, we have $|A| = |B|$. Pair the elements of A and B arbitrarily, say $(a_1, b_1), \dots, (a_m, b_m)$ with $m = |A|$.

Starting from x, for $\ell = 1, \dots, m$ perform the following move: set the coordinate at a_ℓ from 1 to 0 and the coordinate at b_ℓ from 0 to 1. Each such move is exactly of the allowed form in the Markov chain (flipping one 0 to 1 and one 1 to 0), so every intermediate state lies in $\mathcal{X}$ and has exactly I ones.

After performing all m moves, we obtain y, because all coordinates where x and y differ have been corrected. Each of these moves has strictly positive proposal probability and strictly positive acceptance probability since every $p_k \in (0, 1)$. Hence there is a positive-probability path from x to y, so the chain is irreducible on $\mathcal{X}$.

3. *Assume there exist constants $0 < c < C < \infty$ such that*

$$c < \frac{p_k}{1 - p_k} < C \qquad \text{for every } k = 1, \dots, N.$$

Prove that the above algorithm has mixing time $t_{\mathrm{mix}}(\varepsilon) = O(N \log N)$.

Solution. *We use path coupling. Define a graph on $\mathcal{X}$ where $x \sim y$ if they differ in exactly two coordinates (one $0 \to 1$ and one $1 \to 0$), and let*

$$\ell(x, y) = \frac{1}{2} \sum_{k=1}^{N} |x_k - y_k|$$

be the Hamming distance. It suffices to consider neighboring states x, y with $\ell(x, y) = 1$, i.e. there exist unique indices u, v such that

$$x_u = 1,\ x_v = 0, \qquad y_u = 0,\ y_v = 1,$$

and $x_k = y_k$ for all $k \neq u, v$.

At each step, draw the same pair (i_t, j_t) with $i_t \in S_0(x)$ and $j_t \in S_1(x)$ for the x–chain, and couple the y–chain so that it uses the same pair whenever valid (i.e. when $i_t \in S_0(y)$ and $j_t \in S_1(y)$ as well). Since x and y differ only at u, v, we have

$$S_0(x) \triangle S_0(y) = \{u, v\}, \qquad S_1(x) \triangle S_1(y) = \{u, v\}.$$

The only proposal that can make the chains coalesce in one step is $(i_t, j_t) = (v, u)$. This happens with probability $1/((N - I)I)$. For this move the MH ratio is

$$\frac{\pi(x')}{\pi(x)} = \frac{p_v}{1 - p_v} \cdot \frac{1 - p_u}{p_u},$$

which is bounded away from 0 and ∞ by the assumption on $p_k/(1 - p_k)$. In particular there exists $\alpha_0 > 0$ (depending only on c and C) such that the move is accepted in both chains with probability at least α_0. In that event the distance ℓ drops from 1 to 0.

In contrast, if we propose a pair that does not involve both u and v, the distance can either stay at 1 or increase to at most 2. There are at most a constant multiple of N such pairs, each chosen with probability $1/((N-I)I)$, so the possible increase in ℓ is uniformly bounded by some constant $C_1/((N-I)I)$. For N large this is dominated by the negative drift coming from the coalescing proposal. Thus we can find $\kappa > 0$ (depending only on c and C) such that

$$\mathbb{E}[\ell(X_1, Y_1) \mid X_0 = x, Y_0 = y] \le 1 - \frac{\kappa}{N}$$

whenever $\ell(x, y) = 1$.

By the path coupling theorem, for any $x, y \in \mathcal{X}$ and any $t \ge 0$,

$$\mathbb{E}[\ell(X_t, Y_t)] \le \ell(x, y) \left(1 - \frac{\kappa}{N}\right)^t \le N \left(1 - \frac{\kappa}{N}\right)^t \le N e^{-\kappa t/N}.$$

Since

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\mathrm{TV}} \le \mathbb{P}(X_t \ne Y_t) \le \mathbb{E}[\ell(X_t, Y_t)],$$

we obtain

$$\max_{x,y} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\mathrm{TV}} \le N e^{-\kappa t/N}.$$

Solving $N e^{-\kappa t/N} \le \varepsilon$ for t gives

$$t \ge \frac{N}{\kappa} \big(\log N + \log(1/\varepsilon)\big),$$

so the mixing time satisfies $t_{\mathrm{mix}}(\varepsilon) = O(N \log N)$.